

Plusargs and testname

Simulation command lines carry more than the binary, they pass plusargs that configure the run

UVM testname versus custom knobs

- Plus UVM testname is the standard switch for test selection
- If it is missing, you have no selected test from that plusarg
- Seed and verbosity are common UVM knobs, seed for reproducibility, verbosity for log noise
- Custom plusargs like plus num pkts or plus mode are project-specific
- Unknown testname means the factory cannot find the class

Browser lab

Digital Design and Verification Platform

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INTERACTIVE TOOL

Plusargs / CLP

Sketch how **+UVM_TESTNAME** picks the test and other **plusargs** act as run knobs (seed, verbosity, custom). Starter: `+UVM_TESTNAME=base_test +SEED=1` — test selected.

Starter example: `+UVM_TESTNAME=base_test +SEED=1` — known test selected; seed knob applied.

Load starter example

Challenges 0 / 22 cleared

Quiz: `testname`: `+UVM_TESTNAME` primarily...

☐ selects which uvm_test class the run builds

☐ compiles the DUT netlist

☐ replaces the scoreboard

☐ sets the synthesis top

Show hint

Check

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Idle

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Core ideas

+UVM_TESTNAME

Picks which uvm_test class to run.

+SEED

Random seed for replayable runs.

verbosity

Message volume for this invocation.

custom knobs

Project plusargs read via CLP / \$value\$plusargs.

Lab

SCENARIO

COMMAND-LINE PLUSARGS

base_test + seed

+UVM_TESTNAME=base_test +SEED=1

Load preset

Parse

Demo no testname

Explain

Plusargs lab starter

learn_uvm2017 — uvm plusargs

Real UVM literacy

```
wsl - module11-uvm-plusargs/examples/plusargs-sketch

# Track A - real Unix
# real shell session (Track A - UVM plusargs / testName)

$ pwd
/mnt/d/proj/designs/learning/courses/learn_uvm2017/module11-uvm-plusargs

$ ls examples/plusargs-sketch/
cmdline.txt

$ cat examples/plusargs-sketch/cmdline.txt
UVM plusargs sketch (literacy)
UVM plusargs sketch (literacy)

Sim command (example):
./simv +UVM_TESTNAME=base_test +SEED=1 +UVM_VERBOSITY=UVM_LOW \
+NUM_PKTS=16 +MODE=burst

Roles:
+UVM_TESTNAME=<class>  + factory picks which uvm_test to build/run
+SEED=<n>               + randomization reproducibility knob
+UVM_VERBOSITY=<level> + log noise (UVM_LOW / UVM_MEDIUM / UVM_HIGH ...)
+CUSTOM=<value>        + project knobs (NUM_PKTS, MODE, ...)

Reading custom knobs in SV (two common APIs):
// uvm_cmdline_processor (UVM CLP)
string val;
if (uvm_cmdline_proc.get_arg_value("+test_mode", val)) begin
    // use val
end

// SystemVerilog built-in
int n;
if ($value$plusargs("NUM_PKTS=%d", n)) begin
    // use n
end

Makefile pattern:
SIM_ARGS += +UVM_TESTNAME=$(TEST) +SEED=$(SEED)

Common mistakes:
typo in +UVM_TESTNAME + factory cannot find test class
missing + prefix on sim command line
expecting +SEED alone to select a test
custom knob name mismatch between Makefile and get_arg_value string

Test selection vs knobs:
TESTNAME -> which test runs (factory)
everything else -> how the test runs (read inside test/env)

$ grep -n "get_arg_value\[UVM_TESTNAME\|plusarg" ../../learn_uvm2017_sv_verilator/module8/examples/clp/clp.sv | head -12
8: *   3. Understand plusarg handling
28: *   - CLP parsing: uvm_cmdline_proc.get_arg_value()
211: *   - uvm_cmdline_proc.get_arg_value(): Get plusarg value
248:     //   - uvm_cmdline_proc.get_arg_value(): Get plusarg value
274:     void'(uvm_cmdline_proc.get_arg_value("+test_mode", test_mode));
298:     //   - get_arg_value() returns bool (true if found)
311:     if (uvm_cmdline_proc.get_arg_value("+debug_level", str_val)) begin
337:     if (uvm_cmdline_proc.get_arg_value("+num_transactions", str_val)) begin
355:     if (uvm_cmdline_proc.get_arg_value("+seed", str_val)) begin
```

Real shell — plusargs sketch

Pitfalls to watch

- Do not confuse plus UVM testname with a random custom string
- Do not forget the plus prefix on the sim command line
- Do not assume seed alone selects a test
- Typos in testname fail at factory time, not as a missing plusarg
- And remember

Your turn

- Complete the checklist for at least one track, preferably both
- In the browser, parse starter then demo no testname and explain the difference
- On real UVM, write one command line with testname, seed, and a custom knob
- When you are ready